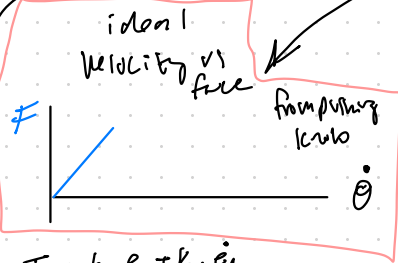
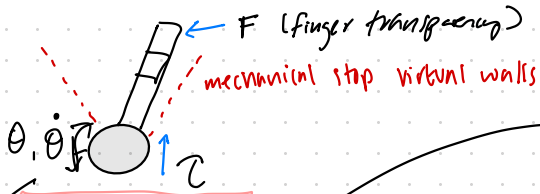


Assuming displacement is angular displacement.

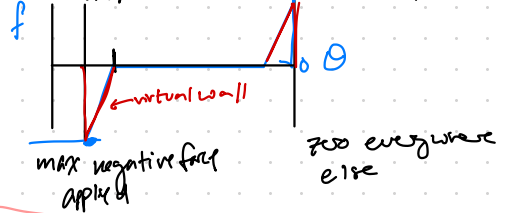
Force Feedback



$$I = k_p e_f + k_d \dot{e}_f$$

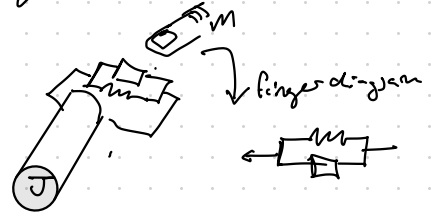
$$e_f = \text{des}_f - \text{act}_f$$

- Want to be transparent; ideal displ vs force



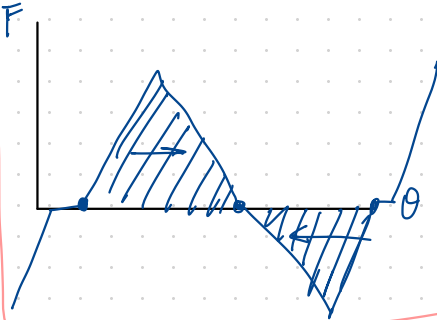
$f = kx$ such that we stay @ zero for as long as possible

finger force diagram

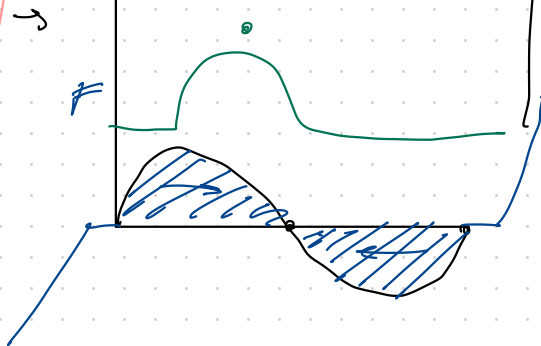


Motion of the system:
 $F = K_s d + k_d \dot{d}$
 $p = \dot{K}x$

local displacement vs. force for collisions



For changing materials / smoother collisions



Torque switch displacement vs. $\Delta \theta$

